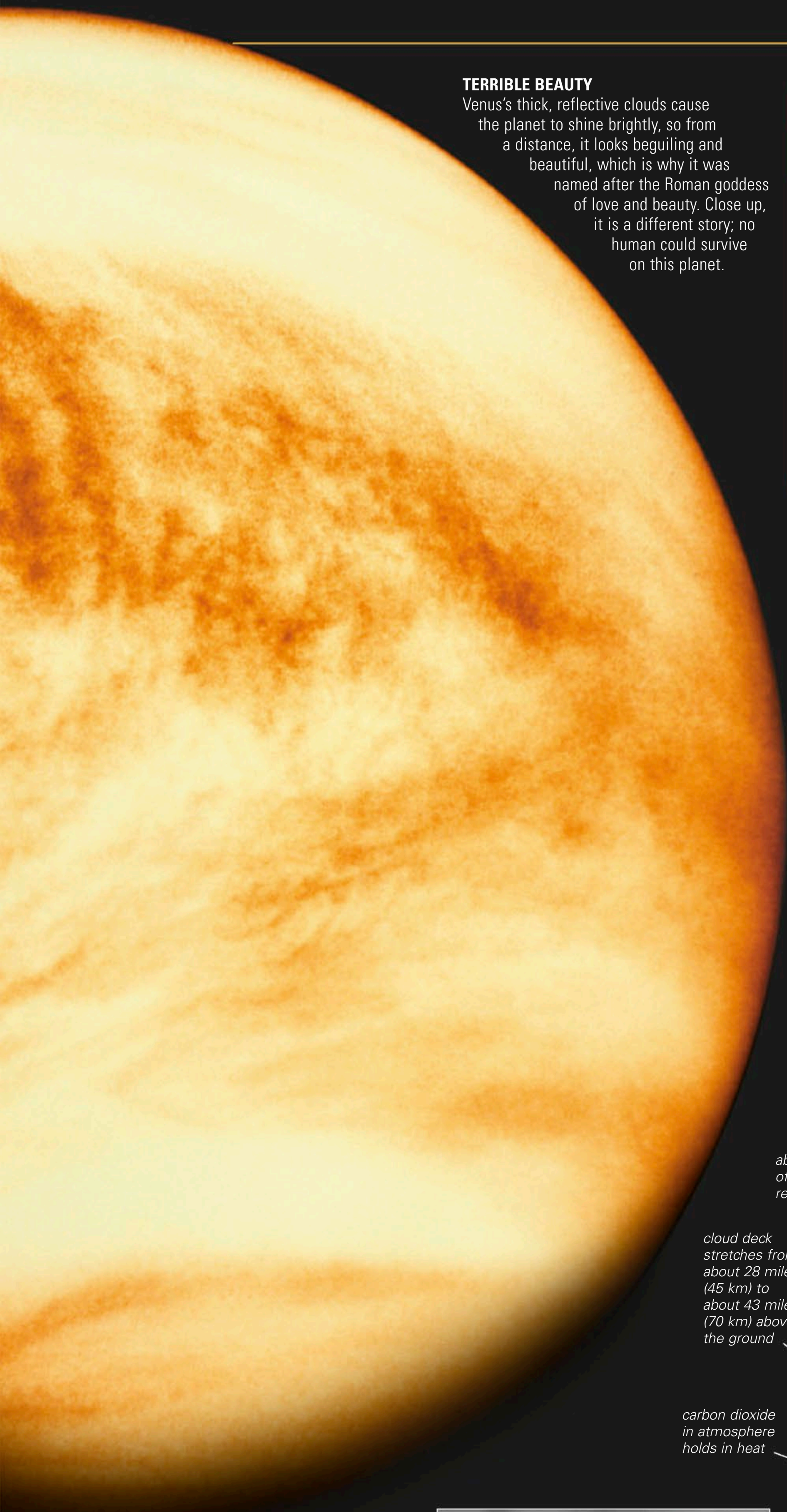
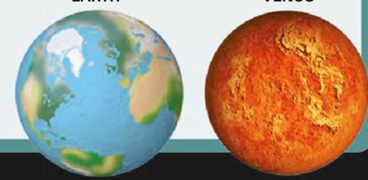
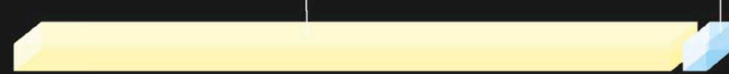


TERRIBLE BEAUTY
Venus's thick, reflective clouds cause the planet to shine brightly, so from a distance, it looks beguiling and beautiful, which is why it was named after the Roman goddess of love and beauty. Close up, it is a different story; no human could survive on this planet.

VENUS PROFILE

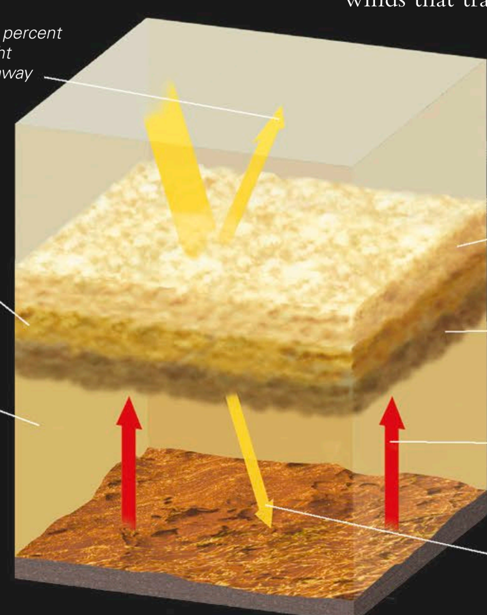
AVERAGE DISTANCE FROM THE SUN 67.2 million miles (108.2 million km)	**ROTATION PERIOD** 243 Earth days
SURFACE TEMPERATURE 867°F (464°C)	**ORBITAL PERIOD (LENGTH OF YEAR)** 224.7 Earth days
DIAMETER 7,521 miles (12,104 km)	**MASS (EARTH = 1)** 0.82
VOLUME (EARTH = 1) 0.86	**GRAVITY AT EQUATOR (EARTH = 1)** 0.9
NUMBER OF MOONS 0	**SIZE COMPARISON**
OBSERVATION Venus is the brightest planet in Earth's sky and is surpassed in brightness only by the Sun and the Moon. Its maximum magnitude is –4.7. It is seen in the early morning or early evening sky.	EARTH VENUS 



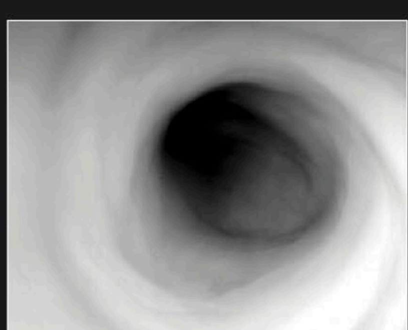
COMPOSITION OF ATMOSPHERE
Along with carbon dioxide and nitrogen, Venus's atmosphere contains traces of other gases, such as water vapor, sulfur dioxide, and argon.

ATMOSPHERE

Venus's carbon-dioxide-rich atmosphere stretches up from the ground for about 50 miles (80 km). A deck of clouds with three distinct layers lies within that atmosphere. The lowest layer is the densest and contains large droplets of sulfuric acid. The middle layer contains fewer droplets, and the top layer has small droplets. Close to the planet's surface, the atmosphere moves very slowly and turns with the planet's spin. Higher up, in the cloudy part of the atmosphere, fierce winds blow westward. The clouds speed around Venus once every four Earth days. The clouds reflect the majority of sunlight reaching Venus back into space, so this is an overcast, orange-colored world. Venus's equator receives more solar heat than its polar regions. Yet the surface temperature at the equator and the poles varies by only a few degrees from 867°F (464°C), as do the day and night temperatures. The initial difference generates cloudtop winds that transfer the heat to the polar regions in one large circulation cell. As a result, Venus has no weather.



GREENHOUSE EFFECT
Venus's thick cloud layers trap heat and help produce the planet's high surface temperature in the same way that glass traps heat in a greenhouse. Only 20 percent of sunlight reaches the surface. Once there, it warms up the rock. Heat in the form of infrared radiation is then released, but it cannot escape and adds to the warming process.



SOUTH POLAR VORTEX
This infrared view from the Venus Express orbiter shows a rotating vortex the size of Europe above the planet's south pole. We see deeper into the cloudy atmosphere near the dark center of this giant storm system. It is known to have persisted for decades, but it sometimes splits into two adjacent whorls.